



PHENIKAA UNIVERSITY
Faculty of Computer Science

Software Analysis and Design

Module 6.2: Unify Analysis Classes

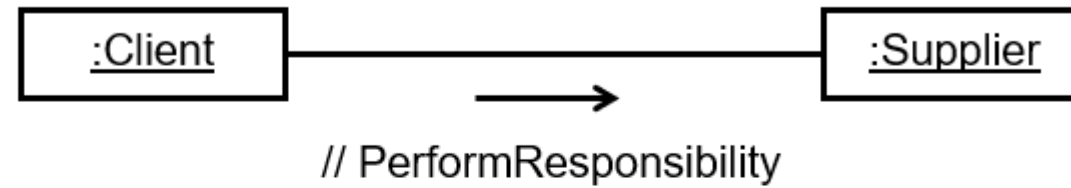
- ❖ Responsibility of analysis class
- ❖ Attributes and Associations of analysis class
- ❖ Unify Analysis Classes
- ❖ Checkpoints

Describe Responsibilities

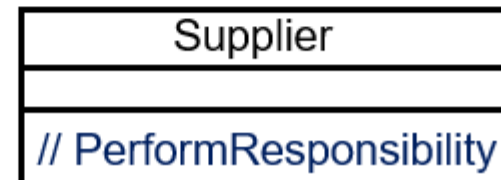


- What are responsibilities?
- How do I find them?

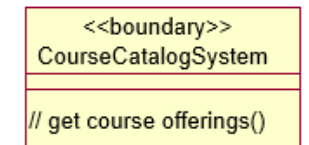
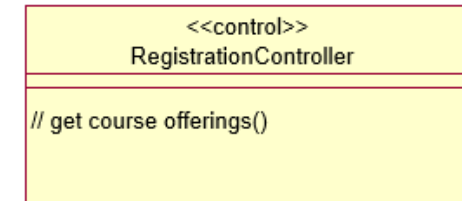
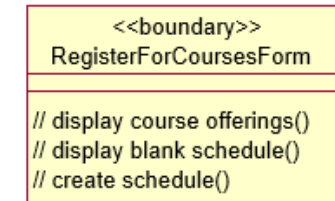
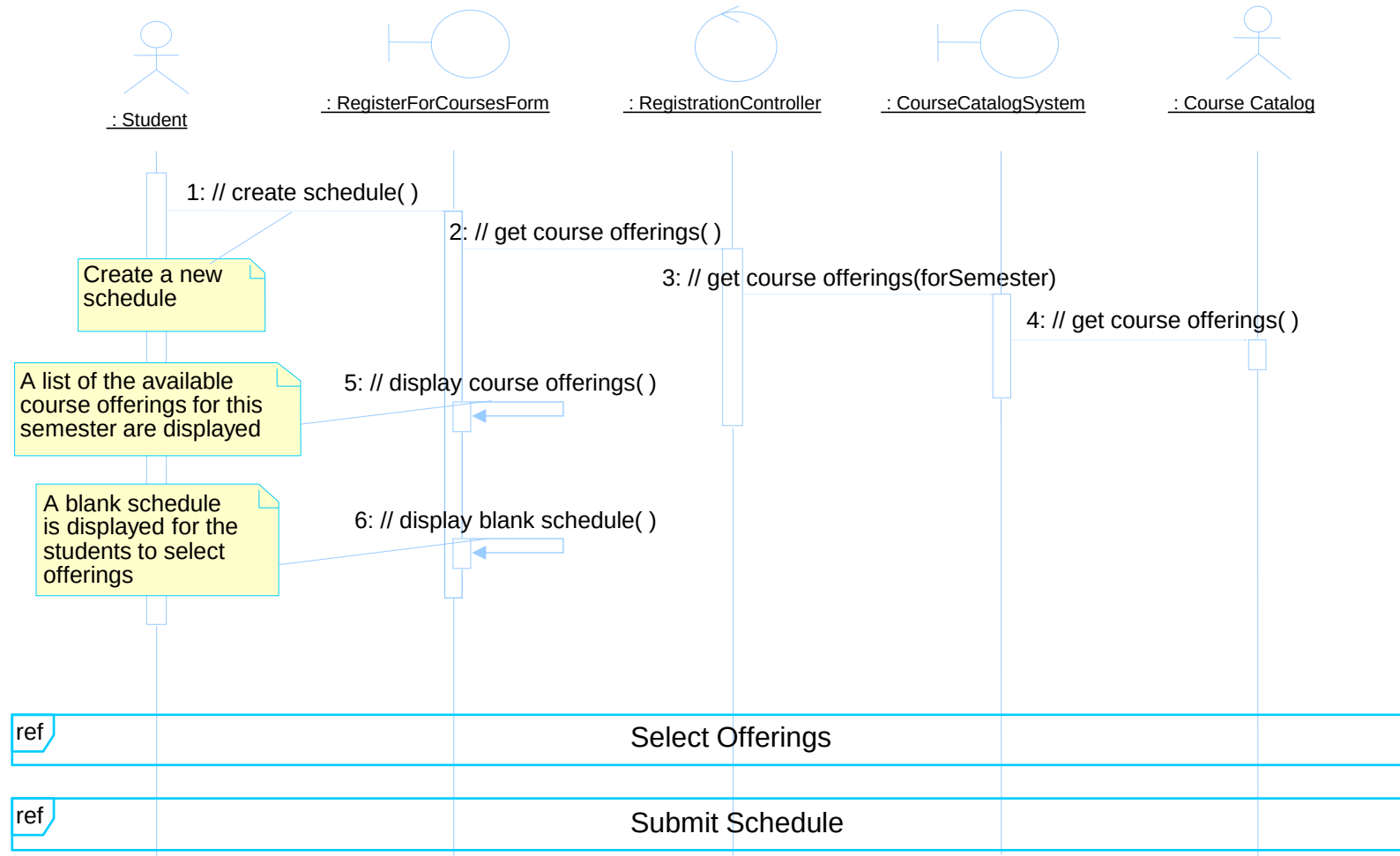
Interaction Diagram



Class Diagram



Example: Sequence Diagrams



Maintaining Consistency: What to Look For



- ❖ In order of criticality
 - ❖ Redundant responsibilities across classes
 - ❖ Disjoint responsibilities within classes
 - ❖ Class with one responsibility
 - ❖ Class with no responsibilities
 - ❖ Better distribution of behavior
 - ❖ Class that interacts with many other classes

- ❖ Responsibility of analysis class
- ❖ Attributes and Associations of analysis class
- ❖ Unify Analysis Classes
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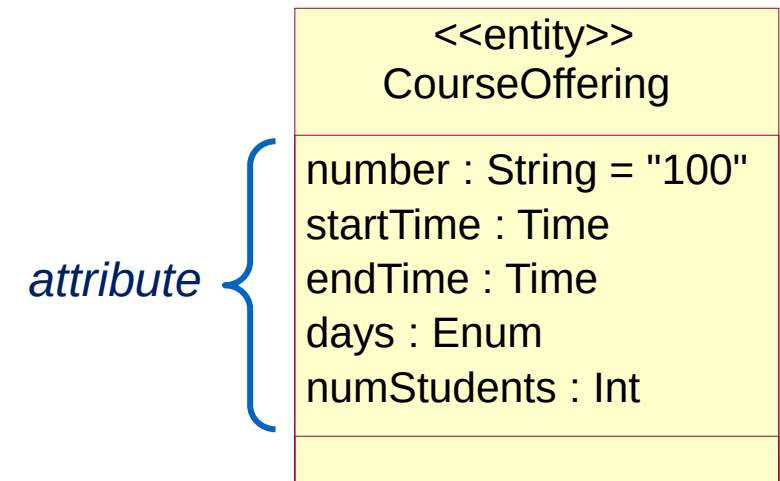
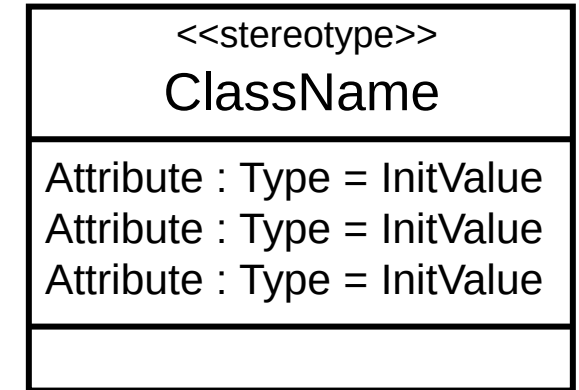
Finding Attributes



❖ Attributes are used to store information. They are atomic things with no responsibilities.

❖ Finding Attributes

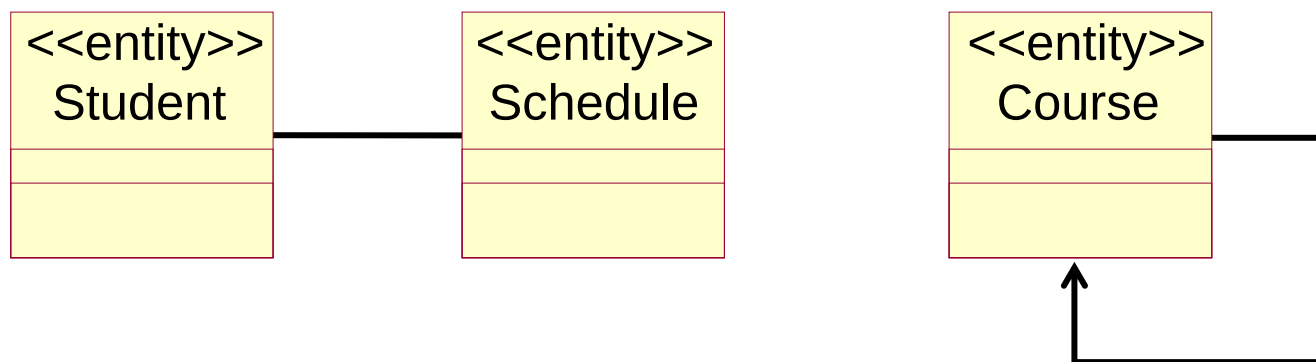
- ❖ Properties/characteristics of identified classes
- ❖ Information retained by identified classes
- ❖ “Nouns” that did not become classes
 - ❖ Information whose value is the important thing
 - ❖ Information that is uniquely "owned" by an object
 - ❖ Information that has no behavior



Review: What Is an Association?



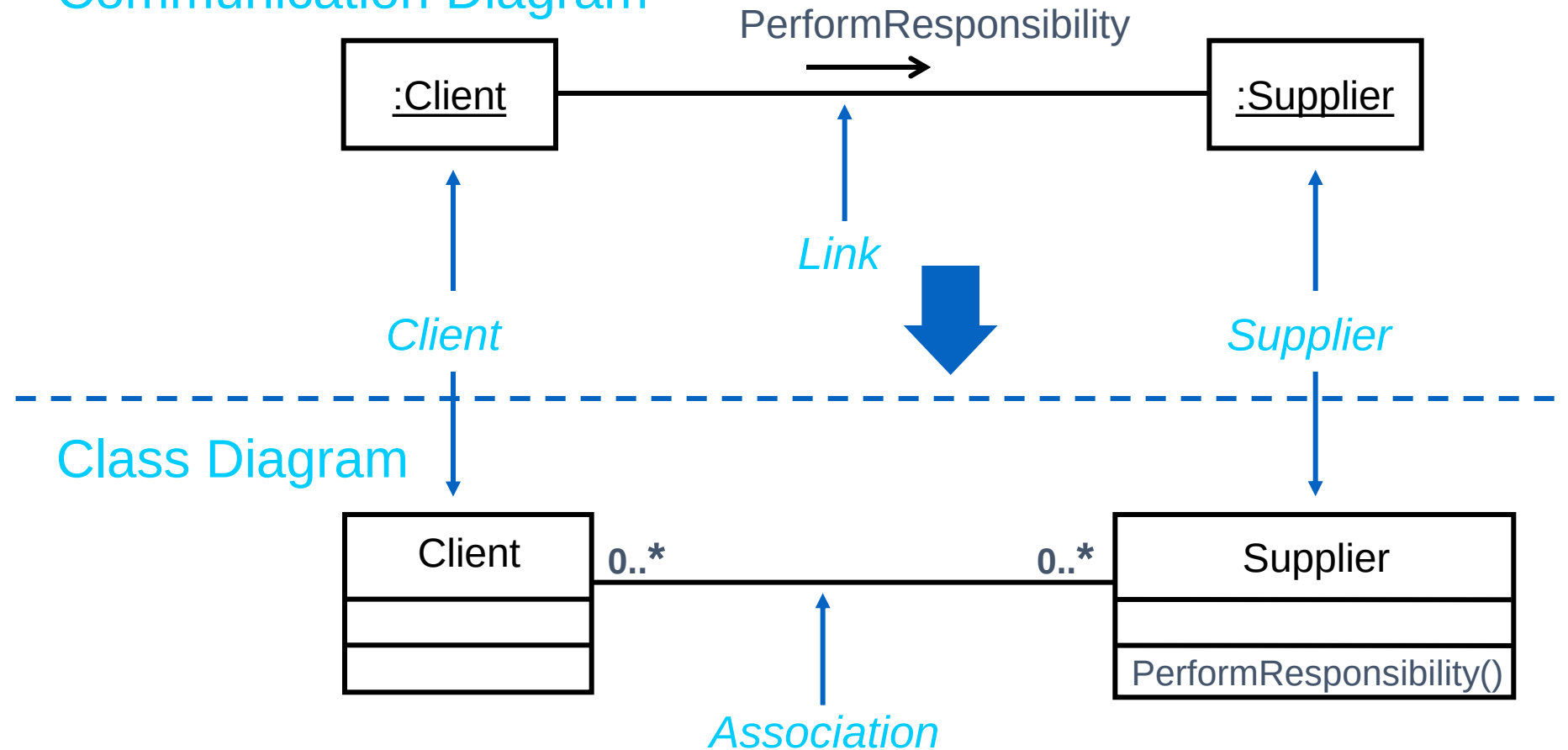
- ❖ The semantic relationship between two or more classifiers that specifies connections among their instances
- ❖ A structural relationship, specifying that objects of one thing are connected to objects of another



Finding Relationships



Communication Diagram

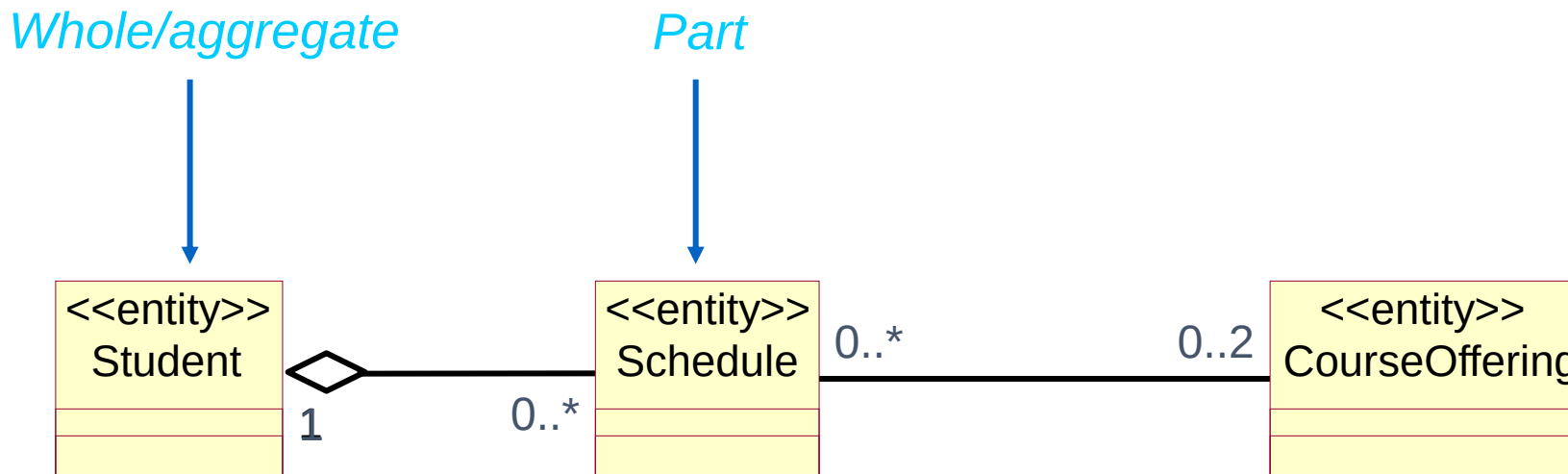


Relationship for every link!

Review: What Is Aggregation?



- ❖ A special form of association that models a whole-part relationship between an aggregate (the whole) and its parts

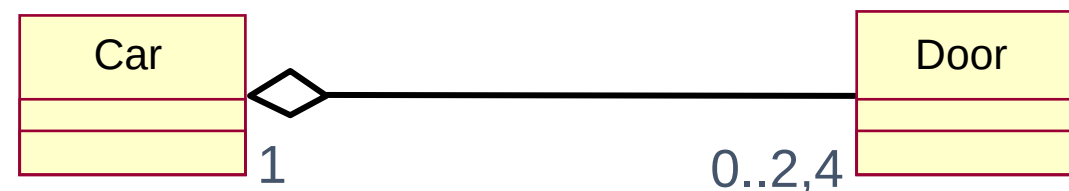


Association or Aggregation?



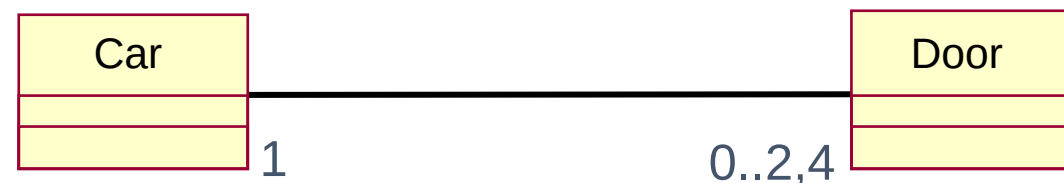
❖ If two objects are tightly bound by a whole-part relationship

❖ The relationship is an aggregation.



❖ If two objects are usually considered as independent, although they are often linked

❖ The relationship is an association.

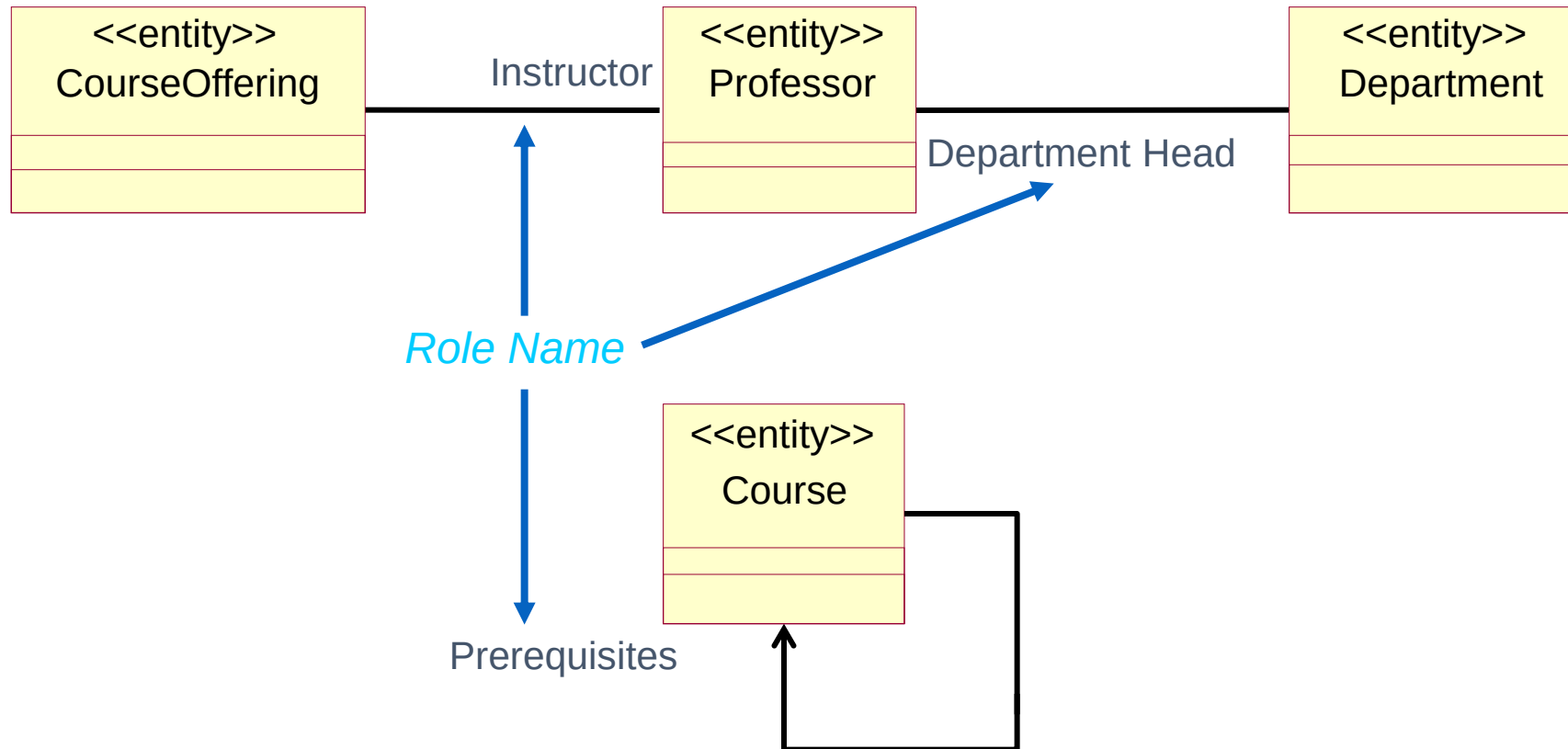


When in doubt, use association.

What Are Roles?



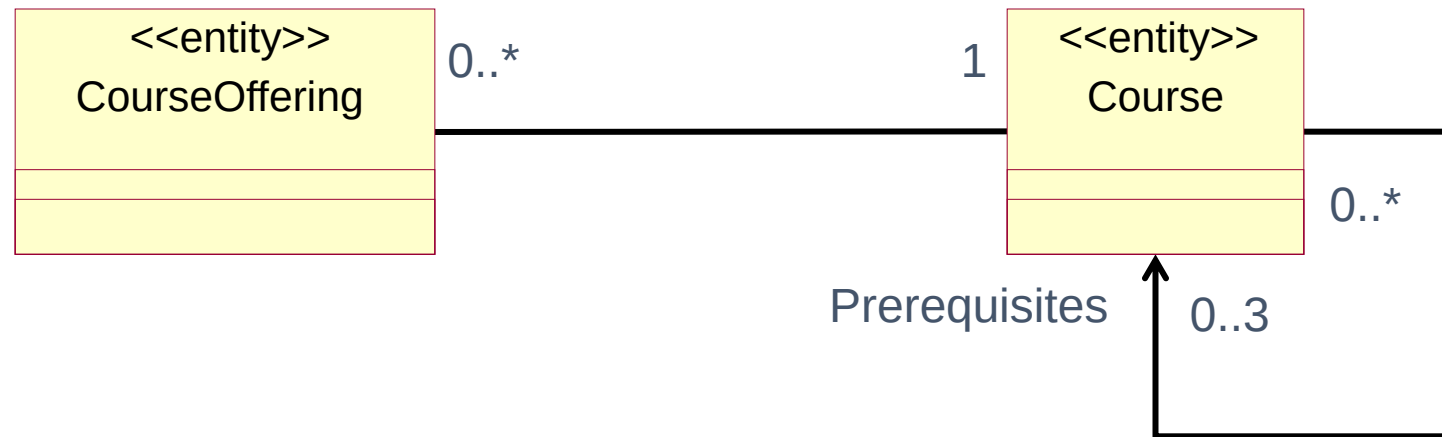
- ❖ The “face” that a class plays in the association



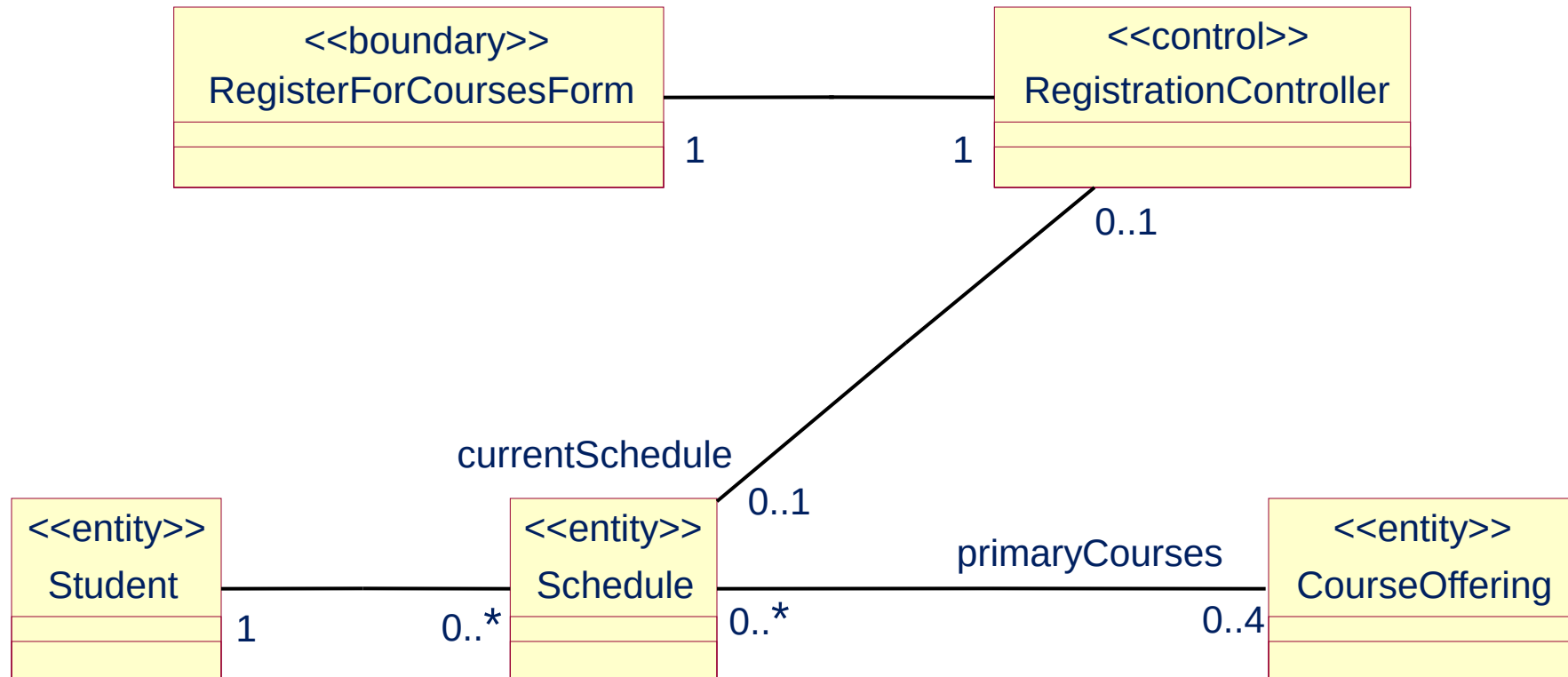
What Does Multiplicity Mean?



- ❖ Multiplicity answers two questions:
 - ❖ Is the association mandatory or optional?
 - ❖ What is the minimum and maximum number of instances that can be linked to one instance?

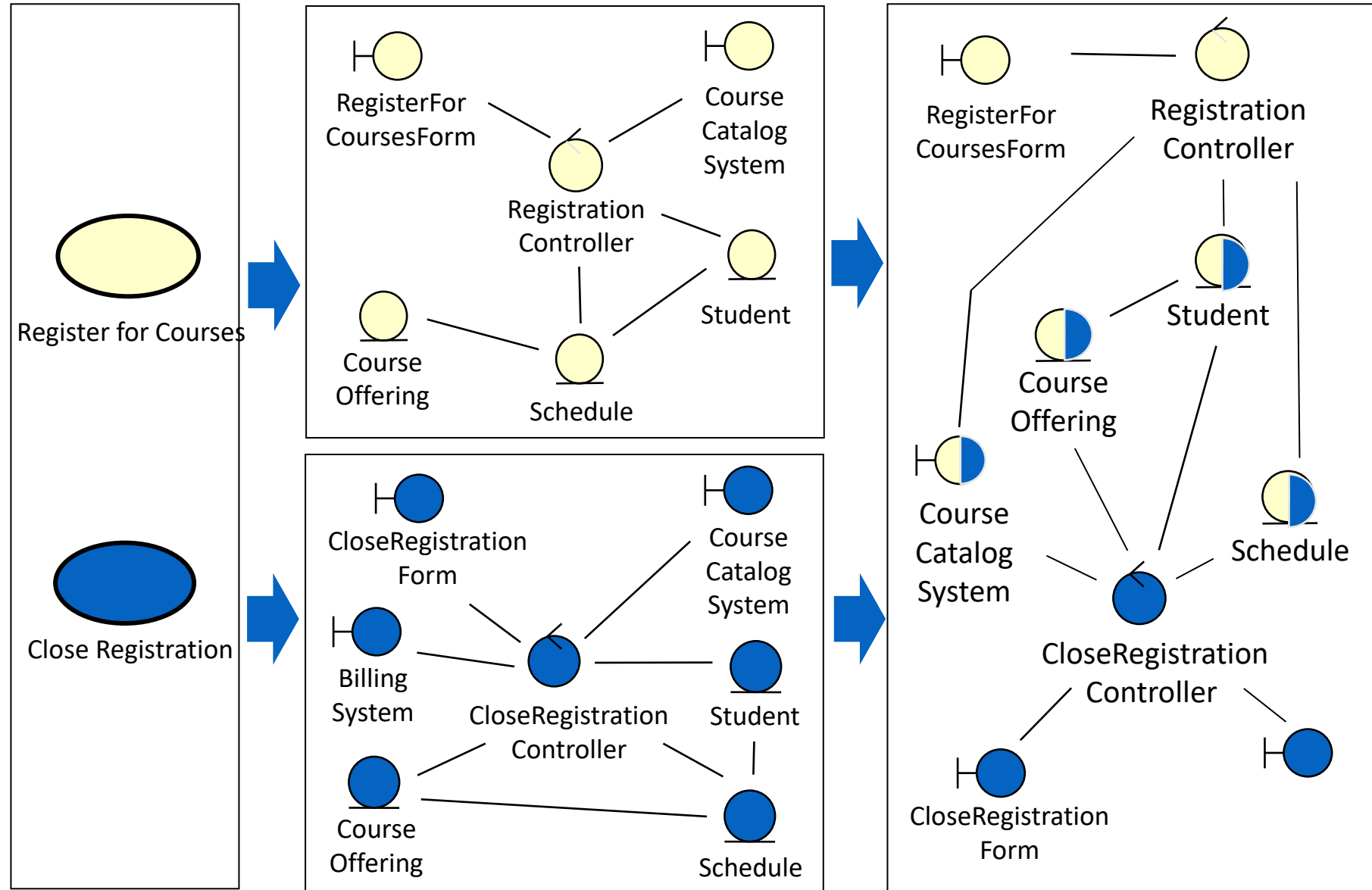


Example: Finding Relationships

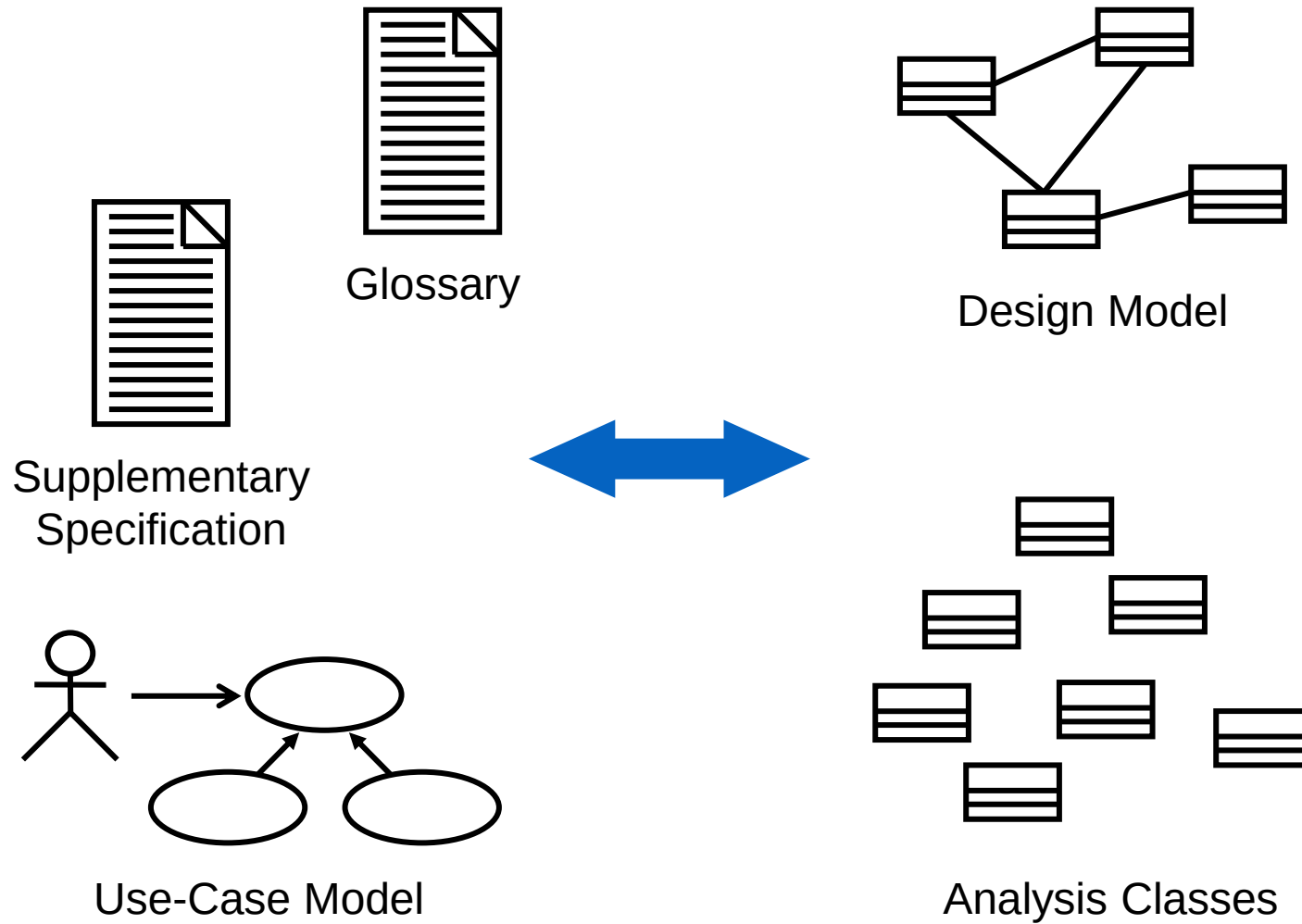


- ❖ Responsibility of analysis class
- ❖ Attributes and Associations of analysis class
- ❖ **Unify Analysis Classes**
- ❖ Checkpoints

Unify Analysis Classes (Hợp nhất lớp phân tích)



Evaluate Your Results



- ❖ Responsibility of analysis class
- ❖ Attributes and Associations of analysis class
- ❖ Unify Analysis Classes
- ❖ Checkpoints

Checkpoints: Analysis Classes



- ❖ Are the classes reasonable?
- ❖ Does the name of each class clearly reflect the role it plays?
- ❖ Does the class represent a single well-defined abstraction?
- ❖ Are all attributes and responsibilities functionally coupled?
- ❖ Does the class offer the required behavior?
- ❖ Are all specific requirements on the class addressed?

Checkpoints: Use-Case Realizations



- ❖ Have all the main and/or sub-flows been handled, including exceptional cases?
- ❖ Have all the required objects been found?
- ❖ Has all behavior been unambiguously distributed to the participating objects?
- ❖ Has behavior been distributed to the right objects?
- ❖ Where there are several Interaction diagrams, are their relationships clear and consistent?

Review: Use-Case Analysis



- ❖ What is the purpose of Use-Case Analysis?
- ❖ What is a Use-Case Realization?
- ❖ What is an analysis class? Name and describe the three analysis stereotypes.
- ❖ Describe some considerations when allocating responsibilities to analysis classes.
- ❖ What two questions does multiplicity answer?



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Q&A